Look Up! A Lesson about the Sky

Day and night, the Sun, shadows, the colour of the sky, stars and the phases of the Moon

SAFETY AND CARE

- **Never look straight at the Sun**, not even for a few seconds, and not through a phone camera, binoculars or a telescope. Say this at the start and again before any outdoor activity.
- Shadow work outdoors: wear a hat, stay in the shade between turns, and bring water.

1. The plan in one look

Min.	Part	Points
15	**A. The sky by day and night**	1, 2
20	**B. The Sun and shadows**	3, 4, 5, 6
15	**C. The colours of the sky**	7
20	**D. Stars**	8, 9, 10
20	**E. The Moon**	11, 12

2. Teaching notes, point by point

For every point, ask first, let several children answer, and then explain. Praise good guesses, even wrong ones: a wrong guess shows how the child is thinking.

1. Have you looked at the sky? What did you see?
Sun, clouds, birds, blue sky. Let every child say one thing.

2. How is the sky different by day and by night?
Blue by day and black at night. Stars are there at night, but we cannot see them by day.

3. Does the Sun move across the sky? How?
It looks that way, because the Earth spins (rotates) once a day. The Sun rises in the east, is highest at noon, and sets in the west. Long ago people thought the Sun went round the Earth; later we learned it is the Earth that spins and travels round the Sun.

4. The Sun makes its own light. The Moon does not.
The Sun is a burning star. The Moon only reflects sunlight, like a mirror.

5. Why should we never look straight at the Sun?
Sunlight is so strong that it can hurt the back of the eye, and the harm can be permanent. Sunglasses are not enough for staring at the Sun.

6. How can shadows tell us where the Sun is?
A shadow always falls on the side away from the Sun. Use a phone torch to show it.

7. Why is the sky blue by day, red at sunset and dark at night?
The sky has no colour of its own. Sunlight is a mix of all colours. The air spreads (scatters) blue light everywhere, so the sky looks blue. At sunset the light has to cross much more air, the blue is scattered away on the way, and red and orange are left.

8. What can we see in the night sky?
Stars, the Moon, and sometimes planets, such as Venus, which shines steady and bright.

9. What are the blinking lights, the stars? Are they like the Sun?
Stars are huge balls of burning gas, far, far away. The Sun is a star too, but it is much closer, so it looks big. Stars seem to twinkle because their light passes through moving air.

10. Where do the stars go in the daytime?
They do not go anywhere. They are still there. Sunlight lights up the air so brightly that we cannot see their faint light. Ask the children for ideas first.

11. Why does the Moon change shape? (full, half, crescent, none)
The Moon is always half lit by the Sun. As the Moon travels round the Earth, we see a different amount of its lit half. (It is not the Earth's shadow.)

12. Can we see the Moon at "new moon"? When can we see it again?
At new moon the lit half faces away from us, so we cannot see it. About a day or two later a thin crescent appears low in the west just after sunset. Ask the children first.



Figure 1. The same hill by day and by night.

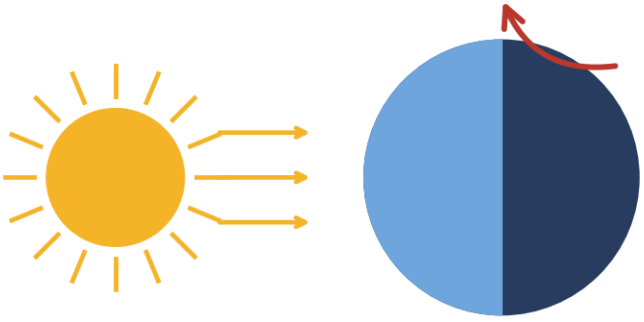


Figure 2. The Earth spins once a day, so the Sun seems to cross our sky. Where we face the Sun it is day; on the other side it is night.

Demonstration 1: the Earth spins (point 3)

Put a lamp or a torch on a table to be the Sun. Ask a child to stand in front of it and turn slowly anti-clockwise on the spot. When she faces the lamp, it is day for her; when her back is to the lamp, it is night. Ask: "Is the lamp moving?" (No: we are turning.)

Demonstration 2: shadows (point 6)

Stand a pencil on clay (or use a child). Shine the phone torch from the left: the shadow falls to the right. Raise the torch higher: the shadow gets shorter. Bring it low: the shadow gets long. So shadows are long in the morning and evening, and shortest at noon.

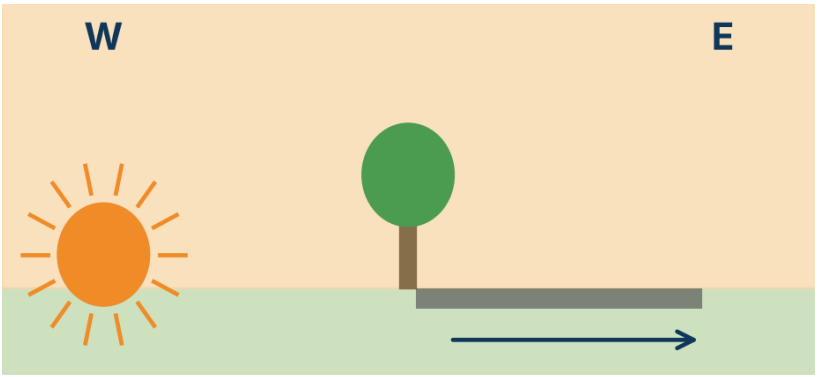


Figure 3. At sunset the Sun is in the west, so shadows point east. Use this to find all the directions.

FOR VOLUNTEERS

Finding the directions (homework 1). At sunset, stand so that your shadow stretches in front of you. You are now facing east. Your back is to the west, your right hand points south and your left hand north. This is only roughly right, because the Sun sets exactly in the west only twice a year (around 21 March and 23 September). In other months it is a little to the north or the south.

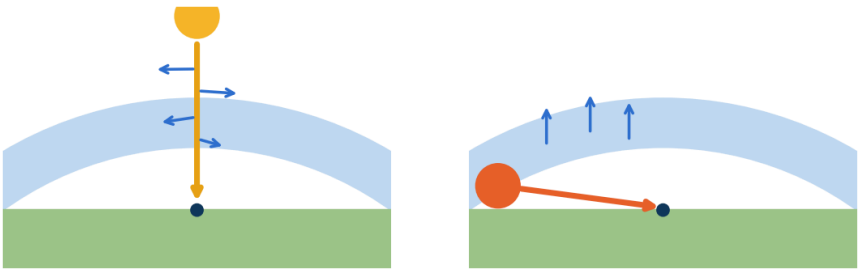


Figure 4. Sunlight is a mix of colours. The air scatters blue light most, so the daytime sky is blue. At sunset the light travels through more air and only red and orange are left.

TIP

Show real photographs of a blue noon sky, a red sunset and a starry night, and ask: "Which colour of the Sun's light is left?" Photos work better than words here.

Demonstration 3: the Moon's phases (points 11 and 12)

In a darkened room put a lamp (the Sun) on one side. A child holds a ball (the Moon) at arm's length and turns slowly all the way round on the spot, as the Moon goes round the Earth. The lamp always lights half of the ball, but the child sees a different amount of the lit half: full when the ball is on the far side from the lamp, none when it is between the child and the lamp.

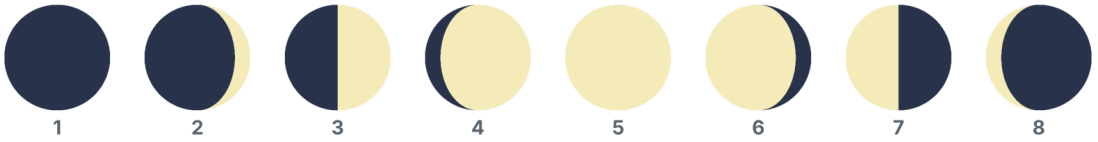


Figure 5. The eight shapes of the Moon over about 29 days: 1 new moon, 2 thin crescent, 3 half (first quarter), 4 more than half, 5 full moon, 6 more than half, 7 half (last quarter), 8 thin crescent. The shapes shown are as seen from India.

KEY IDEA

- The Moon goes through all its shapes in about **29–30 days**. In Bengali we call the bright half of the cycle শুক্লপক্ষ (growing Moon, ending at পূর্ণিমা) and the dark half কৃষ্ণপক্ষ (shrinking Moon, ending at অমাবস্যা).
- Each night the Moon rises about **50 minutes later** than the night before.

ANSWER KEY (VOLUNTEERS ONLY)

- Q10 (stars by day): they are still there; the bright sky hides them.
- Q12 (new moon): we cannot see it; the first thin crescent appears low in the west after sunset, about 1–2 days later. (During a solar eclipse the new moon passes in front of the Sun.)
- Moon phases worksheet: 1 new moon, 2 waxing crescent, 3 first quarter, 4 waxing gibbous, 5 full moon, 6 waning gibbous, 7 last quarter, 8 waning crescent.
- True or false: (1) T; (2) F, the Sun makes its own light; (3) F, never look at the Sun; (4) T; (5) F, the Earth spins; (6) T; (7) F, the Moon has no light of its own; (8) T.

3. Homework and prizes

1. **Directions.** At sunset find out where your shadow falls. It points east. Use it to find north, south and west, and write them on your sheet.
2. **Moon diary.** Every night for one week, look at the Moon, draw its shape and write what you notice. Children who finish get a special prize the next Sunday.

FOR VOLUNTEERS

Some nights are cloudy, and some children cannot see the Moon from home. Tell them that a blank box with "cloudy" written in it also counts as finished. The point is to look, not to be perfect. If the Moon is not visible at all in the week you choose (around new moon), pick a different week.

4. Sky word list

English	Bengali
Sky	আকাশ
Sun	সূর্য
Moon	চাঁদ
Star	তারা
Cloud	মেঘ
Shadow	ছায়া
East / West	পূর্ব / পশ্চিম
North / South	উত্তর / দক্ষিণ
Sunrise / Sunset	সূর্যোদয় / সূর্যাস্ত
Full moon / New moon	পূর্ণিমা / অমাবস্যা

Student worksheets

Name: _____	Date: _____

ক) আকাশে আমি কী কী দেখেছি? এঁকে বা লিখে জানাও।
A) What have I seen in the sky? Draw or write.

খ) দিন আর রাতের আকাশ:
B) The sky by day and by night:

	দিনের আকাশ	রাতের আকাশ
রং কেমন?		
কী কী দেখা যায়?		

গ) ঠিক না ভুল? (✔ বা ✘ দাও)
C) True or false? (✔ or ✘)

	কথা	ঠিক / ভুল
১	সূর্য নিজে আলো দেয়।	
২	চাঁদ নিজে আলো দেয়।	
৩	সরাসরি সূর্যের দিকে তাকানো ভালো।	
৪	সূর্য পূর্ব দিকে ওঠে।	
৫	দিনের বেলা তারারা চলে যায়।	
৬	সূর্যাস্তের সময় ছায়া পূর্ব দিকে পড়ে।	
৭	আকাশের নিজের রং নীল।	
৮	সূর্যও একটি তারা।	

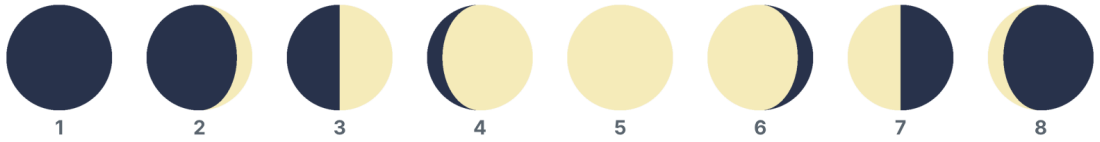
ঘ) ছায়া: সূর্য যেদিকে, ছায়া তার ____ দিকে পড়ে। সকালে আর বিকেলে ছায়া ____ (লম্বা / ছোট), দুপুরে ____ (লম্বা / ছোট)।
D) Shadows: A shadow falls on the ____ side from the Sun. In the morning and evening it is ____ (long / short); at noon it is ____ (long / short).

ঙ) সূর্যাস্তের সময় ছায়া পূর্ব দিকে পড়ে। নীচের ছকে বাকি তিনটে দিক লেখো।
E) At sunset shadows point east. Write the other three directions in the box.

	উত্তর	
পশ্চিম	আমি	পূর্ব (ছায়া এদিকে)
	দক্ষিণ	

(ছকটা নিজের মতো সাজাও: তোমার ছায়া সামনে থাকলে ডান হাত কোন দিকে?)
(Arrange the box to suit you: with your shadow in front, which way is your right hand?)

চ) চাঁদের কলা: ছবিগুলো ১ থেকে ৮ নম্বর। নম্বর দিয়ে নাম মেলাও: অমাবস্যা, পূর্ণিমা, সরু বাঁকা চাঁদ (বাড়তি), সরু বাঁকা চাঁদ (কমতি), অর্ধেক চাঁদ (প্রথম), অর্ধেক চাঁদ (শেষ)।
F) Moon shapes 1 to 8. Match the numbers to the names: new moon, full moon, thin crescent (growing), thin crescent (shrinking), half moon (first), half moon (last).



১	২	৩	৪	৫	৬	৭	৮

Homework 1: directions

সূর্যাস্তের সময় আমার ছায়া এই দিকে পড়েছে: _____ (পূর্ব)। তাহলে আমার ডান হাত _____ দিকে, বাঁ হাত _____ দিকে, পিঠ _____ দিকে।

At sunset my shadow fell towards _____ (east). So my right hand points _____, my left hand _____ and my back _____.

Homework 2: my Moon diary

এক সপ্তাহ রোজ রাতে চাঁদ দেখে আঁকো আর লেখো। মেঘলা থাকলে "মেঘলা" লেখো।

Look at the Moon every night for a week; draw and write. If it is cloudy, write "cloudy".

দিন	তারিখ ও সময়	চাঁদের আকার (আঁকো)	যা লক্ষ্য করলাম
১			
২			
৩			
৪			
৫			
৬			
৭			